USF Bellini
College of Artificial Intelligence, Cybersecurity and Computing

Project 3: Decision Trees (Group Assignment)

Course: Data Structures
Total Points: 100
Due Date: April 17 at 11:59 PM

Objective: You must build a **general tree-based decision tree system** using a **linked structure implementation** (as discussed in Chapter 7).

The program should:

- Read tree data from files
- Construct the tree dynamically
- Analyze tree properties
- Allow interactive exploration

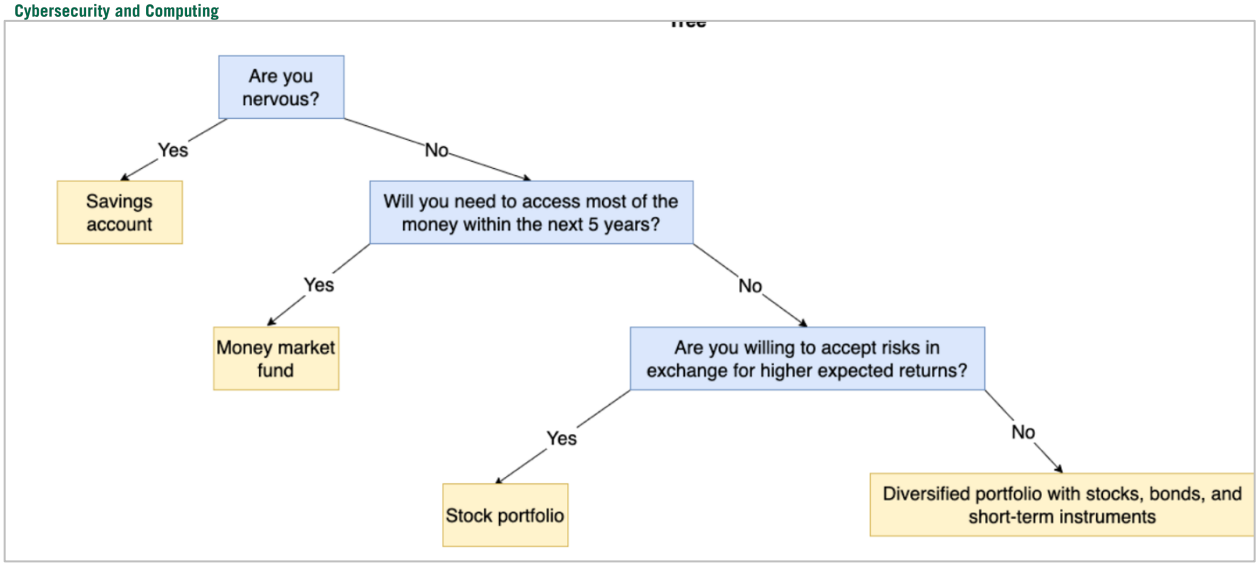
Team Requirements

- This is a **group project**
- **Equal participation is required**
- Work distribution must be documented

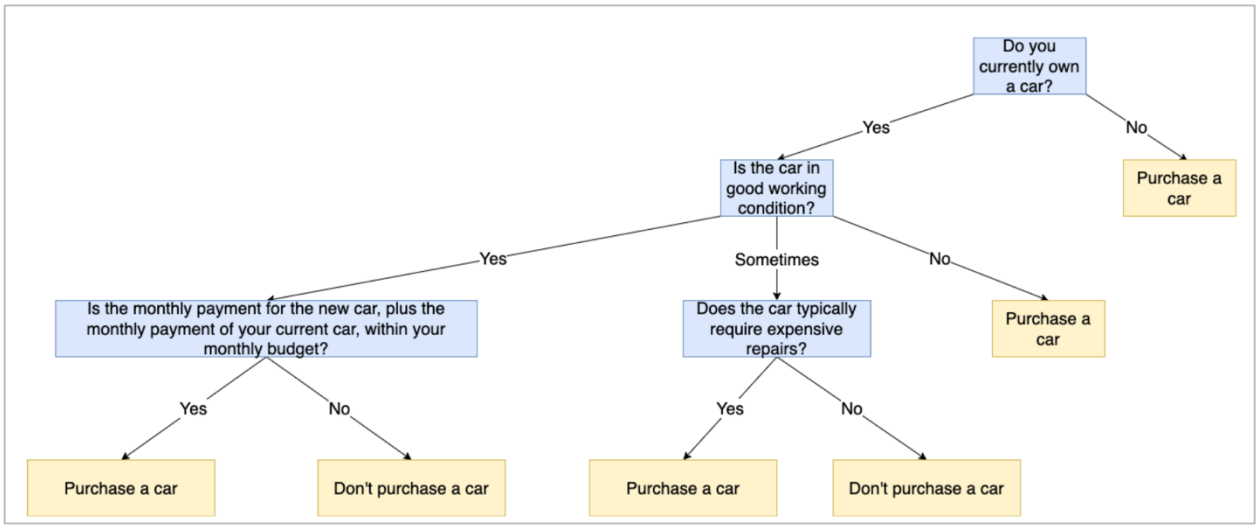
Input Files

You will be given two files:

- tree-car.txt
- tree-investment.txt



1- Investment Advice Decision Tree



2 – Car Purchase Decision Tree

File Format

Each file follows this structure:

1. **First line** → Root node content
2. Each subsequent line contains **4 tab-separated fields**:
    - Level of the node
    - Preorder position of the node

- Edge label (e.g., Yes/No)
- Node content

Core Implementation Requirements

1. Tree Construction

- Build a **LinkedTree data structure**
- Do **NOT hardcode anything**
- Program must work for **any valid input fil**

2. Tree Output File

After building the tree, generate:

File: about_tree.txt

Must include:

A. Tree Visualization

- Use **dash-based indentation**
  - Level n → 2 × n dashes
- Include edge labels in [ ]

Example (from page 8):

--[Yes] Is the car in good working condition?
----[Yes] Purchase a car

B. Tree Properties

Print:

1. **Root node**
2. **Number of internal nodes**
3. **Number of external (leaf) nodes**
4. **Tree height**
5. **Internal nodes (preorder)**
6. **External nodes (preorder)**

C. Binary Tree Analysis

Print:

- Is it a **Binary Tree**? (Yes/No)

If **Yes**, also print:

- Proper Tree (Yes/No)
- Perfect Tree (Yes/No)
- Balanced Tree (Yes/No)

User Interaction (Console Feature)

Your program must allow users to explore the tree:

Prompt:

Which node would you like to explore (enter position or "exit"):

For a valid input:

Display:

- Node content
- One **ancestor**
- One **descendant**
- One **sibling**

Example (page 3):

Node's content: ...
Ancestor: ...
Descendant: ...
Sibling: ...

Invalid Input Handling

- If user enters invalid position:

Invalid input. Please try again.

Exit Condition

- If user types "exit":

Goodbye!

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Submission Requirements

1. PDF Document

Must include:

1. Group member names + group number
2. Instructions to **compile/run on student cluster**
3. Work distribution (percentage per member)
4. Academic integrity statement

2. Code Folder

- Only necessary files
- Proper structure
- Must be zipped as:

<GroupNumber>.zip

Coding Standards

- Use **clear comments** for all functions
- Use **meaningful variable names**
- Use **clean, readable structure**
- Follow **best practices**

Grading Rubric (100 Points)

Criteria	Points
Program runs correctly (cluster instructions)	20
Output correct & matches format	20
Code documentation (comments)	15
Variable & function naming	15
Code quality & readability	10
Correct linked tree implementation	20

Important Constraints

- No hardcoding
- Must handle any valid tree file
- Follow exact output format
- Must match example outputs

What This Project Tests

- Tree data structures (linked representation)
- File parsing
- Recursion / traversal (preorder)
- Tree properties analysis
- Interactive systems design